Reconstructing Plausible Game Histories via Legal-Projected Reverse Generation: A Shogi Case Study

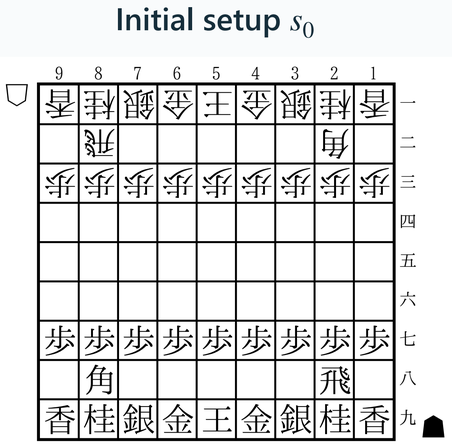
Code & artifacts

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01 One target position, many possible histories

Given only a target position without its move log, reconstruct a legal game history from the start.

Initial setup s_0

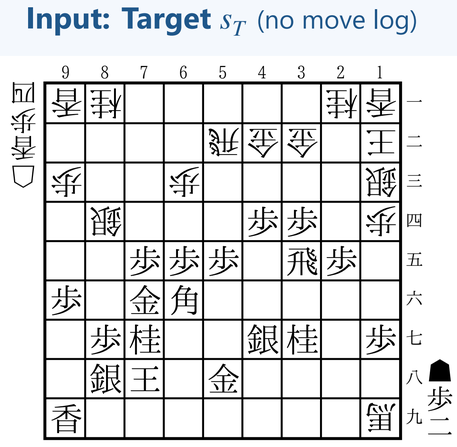


Forward play

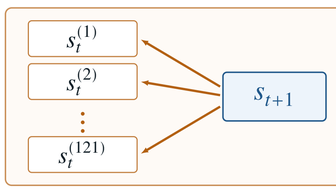
THIS WORK

Reconstruct backward

Input: Target s_T (no move log)



Why much harder to reverse than chess?



121 legal predecessors per step on average

Drops from hand: captured pieces stay in play and can return to empty squares.

Widespread promotion: most piece types promote, multiplying ways to undo moves.

02 96-token reverse Transformer

Input state s_{t+1} (96 discrete tokens):

- 81 Board (9×9)
- 7 Black Hand
- 7 White Hand
- 1 Turn

Bidirectional Transformer Encoder

5.5M params (6 layers, $d = 256$); trained on 102M (s_{t+1}, s_t) pairs

Token cross-entropy + soft piece-count & step-size losses

- 81 Board (9×9) — 29 piece/empty probs
- 7 Black Hand — count probs
- 7 White Hand — count probs
- 1 Turn — flipped

Output $\hat{s}_t$ (96 slots): per-token distributions $p_\theta(\hat{s}_{t,i} \mid s_{t+1}, i)$

Why not direct argmax? **85.5%** of argmax states look statically valid, but only **76.5%** connect by a legal move, breaking multi-step rollback.

03 Enumerate, rank, and step backward

1. Legal predecessors of s_{t+1}

Enumerate positions that legally reach s_{t+1} in one move.

2. Learned score from p_θ

$\sum_{i \in I(c, s_{t+1})} \log p_\theta(c_i \mid s_{t+1}, i)$

Changed tokens $I(c, s_{t+1})$ only;

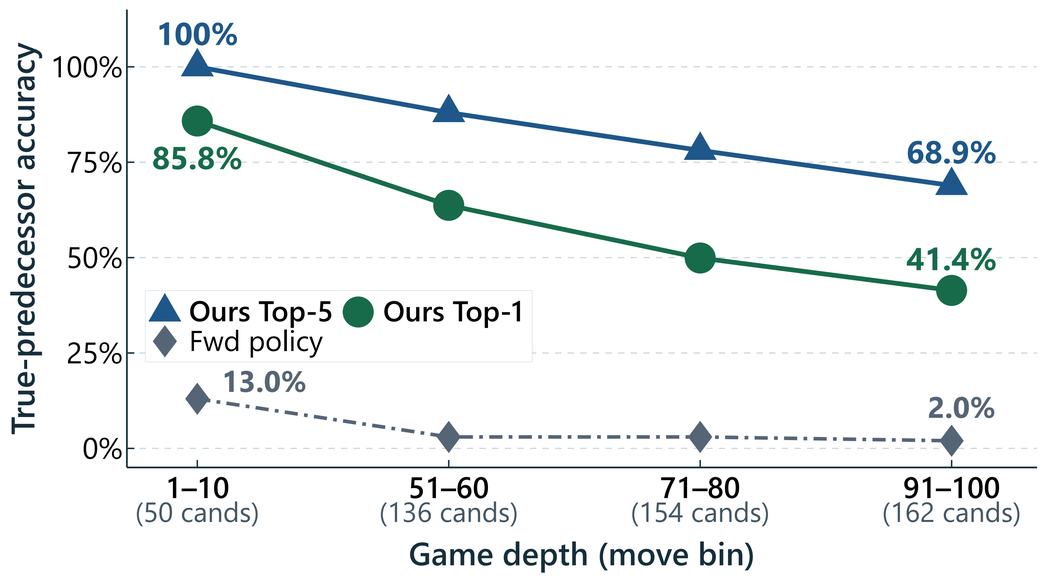
all candidates reuse one forward pass

3. Keep top $k = 10$; step $s_{t+1} \to s_t$

Iterative widening: widths 1–100 (300 for $T \geq 80$), comparing equal depths and reusing cached states.

Distance pruning: discard states whose admissible lower bound $h(s)$ to s_0 exceeds steps left.

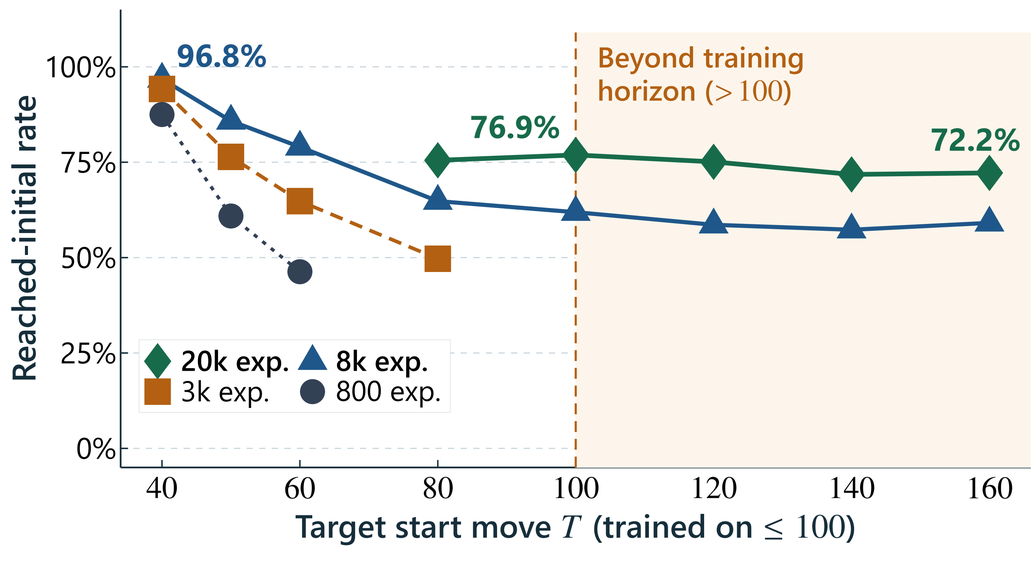
04 Ranking the recorded predecessor



96,023 held-out states (forward-play policy: 100/bin).

Rule-order and minimum-change heuristics: $\leq 2.9\%$ top-1.

05 Rollback beyond the training horizon



Controls score 0%: random, min-change (60, 80 at 8k); forward policy (10–80, first 100 games).

Budgets count unique expanded states ($n = 198$–504 games/depth; 20k run for $T \geq 80$).

06 Legal paths can include rare moves

Start move	Mean policy (Gen. / Rec.)	Paths w/ step $p < 10^{-3}$ (Gen. / Rec.)	Exact-step overlap
40	0.460 / 0.472	19% / 5%	21.8%
50	0.449 / 0.485	44% / 13%	18.6%
60	0.415 / 0.482	75% / 17%	13.3%
80	0.372 / 0.481	95% / 23%	5.8%

Budget 8k (94–100 games/depth). **Means/overlap:** reached paths and paired records.

Low-step rates: **Gen.** = reached paths; **Rec.** = all eligible records.

07 Target-first authoring and scope

Authoring	Specify a climax or puzzle state first; custom scorers could rank legal paths for drama, difficulty, or teaching.
Other games	Applies in principle to discrete, fully observed games with enumerable reverse rules (e.g., chess, checkers).
Limitations	Untracked repetition draws; self-play policy ≠ human study; 10–15 min at 20k (0.03–0.05 s/expansion).